

Below is an organized summary of the escape-room ideas according to the four themes:

General Project Idea

The project will be built as an escape-room game in which the agent moves between several different worlds. Each room has a different task, a different level of difficulty, and a different Reinforcement Learning algorithm. The goal in each room is to reach a single final state that allows the agent to move on to the next room.

The selected themes are:

1. Pacman
2. Museum Heist
3. Car Racing
4. Football

Room 1 — Pacman Room

Algorithm: Dynamic Programming

In the first room, the agent is located inside a Pacman-style maze of size 10x10. Its goal is to collect a certain number of points or coins. Only after collecting them does the exit door open.

The environment is fully known in advance, which makes Dynamic Programming suitable for this room.

Room Components

Pacman-style agent

Points / coins to collect

Walls

Slippery cells

Locked exit door

A door that opens only after all points are collected

Room Goal

Collect all points → open the door → reach the exit

Example Rewards

+10 for collecting a point

+100 for exiting the room

- 1 for each step
- 10 for trying to exit before collecting all points
- 5 for slipping on a slippery cell

What This Room Tests

This room tests whether the agent can plan an optimal path when the environment model is known in advance.

Room 2 — Museum Heist

Algorithm: SARSA

In the second room, the agent is a thief in a museum. It must steal a diamond or a key, avoid security cameras and traps, and then escape through the exit door.

The environment is unknown, so the agent must learn from experience. SARSA is suitable here because it is a more cautious algorithm, especially in an environment that contains risks.

Room Components

Thief
Diamond / key
Security cameras
Traps
Slippery cells
Walls
Exit door

Room Goal

Reach the diamond → avoid cameras and traps → reach the exit door

Example Rewards

- +30 for collecting the diamond / key
- +100 for exiting the room
- 1 for each step
- 25 for entering a camera's range
- 20 for entering a trap
- 5 for trying to exit without the diamond

What This Room Tests

This room tests whether the agent learns to choose a safe path, not necessarily the shortest path. It is possible to create two routes: a short but dangerous route versus a longer but safer route.

Room 3 — Racing Track

Algorithm: Q-Learning

In the third room, the agent is a racing car. The goal is to reach the finish line as quickly as possible. The track is built as a 10x10 grid, but it is presented visually as a racing track.

Q-Learning is suitable here because it searches for the path with the highest future value, and therefore can learn to use shortcuts and boosters.

Room Components

Car

Starting point

Finish line

Track boundaries

Boosters

Oil on the road / slippery cells

Mud that slows down the car

Shortcuts

Danger zones

Room Goal

Reach the finish line in the smallest number of steps possible

Example Rewards

+100 for reaching the finish line

+15 for passing over a booster

-1 for each step

-10 for passing through mud

-20 for slipping on oil

-30 for leaving the track

What This Room Tests

This room tests whether the agent can learn a fast and efficient route. It is also possible to compare SARSA and Q-Learning on the same track and see which one is faster and which one is more cautious.

Room 4 — Football Final Shot

Algorithm: Function Approximation / DQN / Linear Approximation

The fourth room is a continuous football room, not based on a grid. The agent is a player with a ball on a 10x10-meter field. The state is defined by:

X, Y, Vx, Vy

Meaning:

X, Y = the player's position on the field

Vx, Vy = the player's direction and velocity

The goal is to avoid defenders, reach the shooting area, and score a goal against the goalkeeper.

Room Components

Player with a ball

Defenders chasing the player

Goalkeeper moving along the goal line

Shooting area

Goal

Out-of-bounds lines

Possible mud areas / wet grass / speed boost areas

Room Goal

Avoid the defenders → reach the shooting area → shoot → score a goal

Possible Actions

Move Up

Move Down

Move Left

Move Right

Stay

Shoot

Or, as a more advanced option:

Shoot Left

Shoot Center

Shoot Right

Example Rewards

- +300 for scoring a goal
- +50 for entering the shooting area
- +20 for successfully avoiding a defender
- 1 for each step
- 50 if a defender steals the ball
- 40 if the goalkeeper saves the shot
- 30 for leaving the field
- 10 for shooting outside the shooting area

What This Room Tests

This room tests learning in a more continuous and complex environment. Here, the agent does not only search for a path, but must also learn movement, timing, avoidance, and the correct moment to shoot.

This is suitable for function approximation because it is not practical to build a regular Q-table for all the continuous values of X and Y.

Summary Table

Room	Theme	Algorithm	Main Task	Main Difficulty
1	Pacman	Dynamic Programming	Collect points and open the door	Known model and slippery cells
2	Museum Heist	SARSA	Steal a diamond and escape	Cautious learning against cameras and traps
3	Car Racing	Q-Learning	Reach the finish line quickly	Shortcuts, boosters, and risks
4	Football	Function Approximation / DQN	Avoid defenders and score a goal	Continuous movement, velocity, goalkeeper, and defenders